Light

GCSE Computer Science — Character Sets (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Character Sets

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A computer stores all data as binary numbers.
What is the purpose of a character set?

- ○ **A.** To compress text files so they use less storage space
- ○ **B.** To map each character to a unique binary code so text can be stored
- ○ **C.** To convert images into text format for editing
- ○ **D.** To encrypt messages so they cannot be read by others

Q2 (1 mark)

Standard ASCII uses 7 bits to represent each character.
How many unique characters can 7-bit ASCII represent?

- A. 7 characters
- B. 64 characters
- C. 128 characters
- D. 256 characters

Q3 (1 mark)

In ASCII, uppercase letters are assigned consecutive codes starting from 65.
What is the ASCII code for the letter 'B'?

- A. 65
- B. 66
- C. 67
- D. 98

Q4 (1 mark)

In ASCII, lowercase letters start at code 97.
What is the ASCII code for the lowercase letter 's'?

- A. 83
- B. 115
- C. 118
- D. 119

Q5 (1 mark)

ASCII includes codes for digit characters as well as letters.
The digit character '0' has ASCII code 48.
What is the ASCII code for the digit character '3'?

- ○ **A.** 3
- ○ **B.** 48
- ○ **C.** 51
- ○ **D.** 99

Q6 (1 mark)

Different computer manufacturers originally used different character codes.
Why is it important that computers use a standardised character set like ASCII?

- ○ **A.** So that text files can be shared between different computer systems
- ○ **B.** So that computers can run faster when processing text
- ○ **C.** So that text files take up less storage space
- ○ **D.** So that images can be stored alongside text

Q7 (1 mark)

In ASCII, uppercase 'M' has the code 77.
What is the ASCII code for lowercase 'm'?

- ○ **A.** 45

- B. 77
- C. 109
- D. 141

Q8 (1 mark)

A computer stores the character 'K' using ASCII.
The ASCII code for 'K' is 75.
What is 75 in 7-bit binary form?

- A. 0100101
- B. 1001011
- C. 1001101
- D. 1110101

Q9 (1 mark)

ASCII codes 0 to 31 are reserved for control characters.
What are control characters used for?

- A. To represent uppercase letters
- B. To represent punctuation marks
- C. To control devices and formatting, such as new lines and tabs
- D. To represent numbers 0 to 31

Q10 (1 mark)

The 128 characters in standard ASCII include several categories.
Which of the following is NOT included in standard ASCII?

- ○ **A.** Uppercase letters A-Z
- ○ **B.** Digit characters 0-9
- ○ **C.** Chinese characters
- ○ **D.** Punctuation marks

Q11 (1 mark)

A text file contains 500 characters.
If each character is stored using 7-bit ASCII, how many bits are needed to store the file?

- ○ **A.** 500 bits
- ○ **B.** 3,500 bits
- ○ **C.** 4,000 bits
- ○ **D.** 64,000 bits

Q12 (1 mark)

Extended ASCII uses 8 bits per character instead of 7 bits.
How many unique characters can extended ASCII represent?

- ○ **A.** 128 characters
- ○ **B.** 192 characters
- ○ **C.** 256 characters

- **D.** 512 characters

Q13 (2 marks)

A programmer stores the word 'BITS' in memory using ASCII.
The ASCII codes are: B=66, I=73, T=84, S=83.
What is the total of all the ASCII codes for this word?

- **A.** 276
- **B.** 294
- **C.** 306
- **D.** 318

Q14 (2 marks)

A program needs to convert uppercase letters to lowercase.
The ASCII code for 'K' is 75.
Which calculation correctly converts 'K' to 'k'?

- **A.** 75 − 32 = 43
- **B.** 75 + 32 = 107
- **C.** 75 × 2 = 150
- **D.** 75 ÷ 2 = 37.5

Q15 (2 marks)

Two text files contain different amounts of text:
• File A: 3,000 characters stored in 7-bit ASCII
• File B: 2,500 characters stored in 8-bit extended ASCII

Which file is larger?

- ○ **A.** File A (21,000 bits) is larger
- ○ **B.** File B (20,000 bits) is larger
- ○ **C.** Both files are the same size
- ○ **D.** File A (24,000 bits) is larger

Q16 (2 marks)

In ASCII, uppercase letters have consecutive codes from A=65 to Z=90.
A program sorts words alphabetically by comparing ASCII codes.
Which comparison is correct?

- ○ **A.** 'CAT' comes before 'BAT' because C (67) > B (66)
- ○ **B.** 'BAT' comes before 'CAT' because B (66) < C (67)
- ○ **C.** 'BAT' comes before 'CAT' because A (65) < A (65)
- ○ **D.** 'CAT' comes before 'BAT' because T (84) < T (84)

Q17 (2 marks)

The ASCII code for the letter 'z' is 122.
What is the binary representation of 'z' in 7-bit ASCII?

- ○ **A.** 1111010
- ○ **B.** 1111011
- ○ **C.** 1111100

○ **D.** 1101010

Q18 (2 marks)

A company creates software for international customers in Japan, Saudi Arabia, and Greece.
Why is standard 7-bit ASCII unsuitable for this software?

○ **A.** ASCII is too slow for modern computers to process

○ **B.** ASCII can only represent 128 characters, which cannot include Japanese, Arabic, and Greek alphabets

○ **C.** ASCII files are too large for international transfer

○ **D.** ASCII was designed only for printing, not for screen display

Q19 (2 marks)

Extended ASCII uses 8 bits and includes 256 characters.
The first 128 characters (codes 0-127) are identical to standard ASCII.
What is the benefit of this design?

○ **A.** Extended ASCII files are smaller than standard ASCII files

○ **B.** Standard ASCII text can be read correctly by extended ASCII systems without conversion

○ **C.** Extended ASCII can represent unlimited characters

○ **D.** Extended ASCII is faster to process than standard ASCII

Q20 (2 marks)

A game developer needs to create a character set for a fantasy game with exactly 64 unique symbols.
How many bits are needed per character to encode all 64 symbols?

- ○ **A.** 5 bits
- ○ **B.** 6 bits
- ○ **C.** 7 bits
- ○ **D.** 8 bits

Q21 (2 marks)

A password 'Kz3' contains mixed case letters and a digit.
Using ASCII: K=75, z=122, '3'=51.
How many bits are needed to store this password in 7-bit ASCII?

- ○ **A.** 3 bits
- ○ **B.** 21 bits
- ○ **C.** 24 bits
- ○ **D.** 248 bits

Q22 (2 marks)

A text document contains 4,500 characters stored in 8-bit extended ASCII.
What is the file size in bytes?
(1 byte = 8 bits)

- ○ **A.** 562.5 bytes
- ○ **B.** 4,500 bytes

- ○ **C.** 36,000 bytes
- ○ **D.** 4,500 kilobytes

Q23 (2 marks)

A text file contains the message 'HELLO' followed by a new line character and then 'WORLD'.
The new line character has ASCII code 10.
How many characters are stored in total?

- ○ **A.** 10 characters
- ○ **B.** 11 characters
- ○ **C.** 12 characters
- ○ **D.** 21 characters

Q24 (3 marks)

A software company designs a custom character set for a specialist application. The set must include:
• 52 letters (uppercase and lowercase)
• 10 digits
• 40 punctuation and symbol characters
• 20 special function codes
What is the minimum number of bits needed per character?

- ○ **A.** 6 bits (64 possible values)
- ○ **B.** 7 bits (128 possible values)
- ○ **C.** 8 bits (256 possible values)

- ○ **D.** 9 bits (512 possible values)

Q25 (3 marks)

A database stores customer names. The administrator compares storage options:
• Option A: 7-bit ASCII, storing 6,000 names averaging 15 characters each
• Option B: 8-bit extended ASCII, storing the same data
How much additional storage does Option B require?

- ○ **A.** 6,000 bits (750 bytes)
- ○ **B.** 15,000 bits (1,875 bytes)
- ○ **C.** 90,000 bits (11,250 bytes)
- ○ **D.** 720,000 bits (90,000 bytes)

Q26 (3 marks)

A student makes these statements about character sets:
1. ASCII uses fewer bits than Unicode, so ASCII files are always smaller
2. Unicode can represent characters from all world languages
3. ASCII codes 65-90 represent the same characters in both ASCII and Unicode
4. Extended ASCII is universally standardised across all computer systems
Which statements are correct?

- ○ **A.** Statements 1 and 4 only
- ○ **B.** Statements 2 and 3 only
- ○ **C.** Statements 1, 2, and 3 only

- ○ **D.** All statements are correct

Q27 (3 marks)

A programmer finds that a text file displays incorrectly on different computers. The letter 'é' (e with accent) appears as 'Ã©' on some systems.
What is the most likely cause of this problem?

- ○ **A.** The file is corrupted and needs to be restored from backup
- ○ **B.** The computers are using different extended ASCII or encoding standards to interpret the same bytes
- ○ **C.** The letter 'é' cannot be stored in any digital format
- ○ **D.** The file was compressed, which removed the accent marks

Q28 (3 marks)

A program converts text to lowercase using this algorithm:
`FOR each character c in text:`

```
  IF ASCII code of c is between 65 and 90 inclusive:
    Add 32 to the ASCII code
  Output the character
```

What is the output when the input is 'Code123'?

- ○ **A.** 'code123'
- ○ **B.** 'CODE123'
- ○ **C.** 'code!#$'

- D. 'cODE123'

Q29 (3 marks)

A network transmits text using 7-bit ASCII. Due to noise, one bit in the character 'M' (ASCII 77, binary 1001101) is flipped from 1 to 0.
The corrupted binary is 1001100.
What character will the receiver display?

- A. The letter 'L' (ASCII 76)
- B. The letter 'N' (ASCII 78)
- C. The letter 'm' (ASCII 109)
- D. An error symbol because the code is invalid

Q30 (3 marks)

A company stores 50 million customer records. Each record has a 'country' field that can contain one of 150 different country names. Currently, country names are stored as text (average 10 characters in 8-bit ASCII).
A developer proposes storing a numeric country code instead.
What is the storage saving per record using the minimum bits needed for 150 countries?

- A. 70 bits saved (80 bits reduced to 10 bits)
- B. 72 bits saved (80 bits reduced to 8 bits)
- C. 73 bits saved (80 bits reduced to 7 bits)
- D. 76 bits saved (80 bits reduced to 4 bits)

Total: **55** marks • Time: **30 min**

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